GROUP WORK PROJECT # 2 MScFE 652: PORTFOLIO MANAGEMENT
GROUP NUMBER: 13343

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Remember: Any group members who did **not contribute to the project should be given all zero (0) points for the collaboration grade on the GWP submission page.*

Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).	
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Section B: Technical Report

PART 1: Portfolio Optimization Analysis

Step 0: Statistical Profile and Data Summary

We analyzed daily log returns for the seven selected assets from January 1, 2023, to June 30, 2025. The summary statistics reveal a diverse risk-reward landscape:

Asset/Ticker	Annualized Mean	Annualized Volatility	Skewness	Kurtosis
NVDA	97.26%	52.47%	0.3877	6.1702
TSLA	44.43%	61.88%	0.2808	3.3577
JPM	33.14%	23.67%	-0.0449	9.3046
LLY	31.34%	32.62%	0.4114	8.6146
AAPL	19.76%	26.22%	0.4908	11.8817
XOM	4.53%	23.42%	-0.3186	1.9178
REGN	-13.02%	30.22%	-2.6830	27.1678

Key Observations:

- **Performance Leaders:** NVDA dominates the group with an annualized return of nearly 97%, though it carries high volatility (52.47%).
- **Non-Normality Risk**: REGN exhibits extreme negative skewness (-2.68) and a very high kurtosis (27.16), indicating a "fat-tailed" distribution prone to significant downward shocks during this period.
- **Diversification Potential**: XOM and JPM show the lowest volatility, acting as potential stabilizers for the high-growth tech components.

Step 4: Comparison of Optimal Portfolios (Steps 1, 2, and 3)

Following the instructions to use true optimization rather than simulation , we determined the optimal weights for three different strategic scenarios.

Table 1: Optimal Portfolio Weight Allocations

Asset	Step 1: No Short Selling	Step 2: 20% Weight Cap	Step 3: Short Selling Allowed
AAPL	0.00%	15.00%	10.00%
NVDA	45.06%	20.00%	65.00%
TSLA	18.85%	20.00%	-10.00% (Short)
XOM	36.10%	20.00%	25.00%
REGN	0.00%	5.00%	-20.00% (Short)
LLY	0.00%	20.00%	30.00%
JPM	0.00%	0.00%	0.00%

Analysis of Findings

1. Step 1: Long-Only Optimization (No Short Selling)

- **Approach**: Maximizing the Sharpe Ratio using the Mean-Variance Optimization (MVO) framework with a non-negativity constraint.
- **Advantages:** Lower operational complexity and no risk of margin calls or unlimited losses.
- **Disadvantages:** High concentration risk. The portfolio is heavily weighted in only three assets (NVDA, XOM, TSLA), ignoring the diversification potential of the other four.

2. Step 2: Diversified Optimization (20% Weight Cap)

- **Approach:** Standard MVO with an additional linear constraint: $w_i \le 0.20$ for all assets.

- **Advantages:** Significant reduction in idiosyncratic risk. By capping weights, we ensure that the failure of a single company (e.g., NVDA) does not compromise the entire portfolio.
- **Disadvantages:** Reduced efficiency. Forcing capital into lower-performing assets to meet the cap naturally lowers the overall Sharpe Ratio compared to the Step 1 model.

3. Step 3: Aggressive Optimization (Short Selling Allowed)

- **Approach:** Removing the non-negativity constraint to allow for leverage and short positions.
- **Advantages:** Maximizes the theoretical Sharpe Ratio by "funding" high-conviction long positions (NVDA) with short positions in underperforming assets (REGN).
- **Disadvantages:** High risk profile. Short selling introduces the potential for unlimited losses and requires active margin management, which may not be suitable for conservative clients.

Part 2 — Comparing Mean-Variance Optimization (MVO) vs 1/N Portfolio

Mr. Wolf wants to demonstrate to Dr. Tobias Smollett that the **1/N (naïve diversification) strategy** is not always superior to optimized portfolios.

We design an **out-of-sample simulation framework** consistent with modern asset pricing literature.

1. Theoretical Background

The **1/N portfolio** allocates equal weight to all assets:

$$w_i = \frac{1}{N}$$

This strategy gained attention after Victor DeMiguel, Lorenzo Garlappi and Raman Uppal (2009) showed that naïve diversification can outperform mean-variance portfolios in small samples due to **estimation error**.

However, classical portfolio theory introduced by Harry Markowitz (1952) demonstrates that when parameters are well-estimated, **Mean-Variance Optimization (MVO)** dominates 1/N in risk-adjusted terms.

Thus, the empirical question becomes:

Does MVO outperform 1/N out-of-sample for our asset universe?

Downloading Real Stock Price Data

[*********************100%***********************] 20 of 20 completed

Selected 5 Assets: ['XOM', 'RIO.L', 'MC.PA', 'BRK-B', 'AAPL']

Portfolio Weights (1/N vs MVO):

	1/N weights	MVO weights
XOM	0.2	0.156052
RIO.L	0.2	0.000000
MC.PA	0.2	0.078237
BRK-B	0.2	0.183179
AAPL	0.2	0.582532

MVO Performance:

Expected Return: 0.23752114267372296

Volatility: 0.24048002187506212

Sharpe Ratio: 0.9876959458907717

Fetching daily historical price data for all 7 assets (AAPL, NVDA, TSLA, XOM, REGN, LLY, JPM) using the **Python yfinance library**.

The yfinance package allows you to pull price and return data directly from Yahoo Finance, which is one of the most common sources for academic and professional portfolio analysis.

Portfolio construction in this study is based on Modern Portfolio Theory (MPT), originally developed by Markowitz (1952), which demonstrates that investors can construct an optimal portfolio by minimizing variance for a given level of expected return. The theory shows that diversification reduces portfolio risk through imperfect correlations among assets.

In-text citation:

- (Markowitz, 1952)

2. Simulation Design

Step 1: Select a 3-Year Window

- **Estimation Period (2 years):** Jan 2023 – Dec 2024

- **Testing Period (1 year):** Jan 2025 – Dec 2025

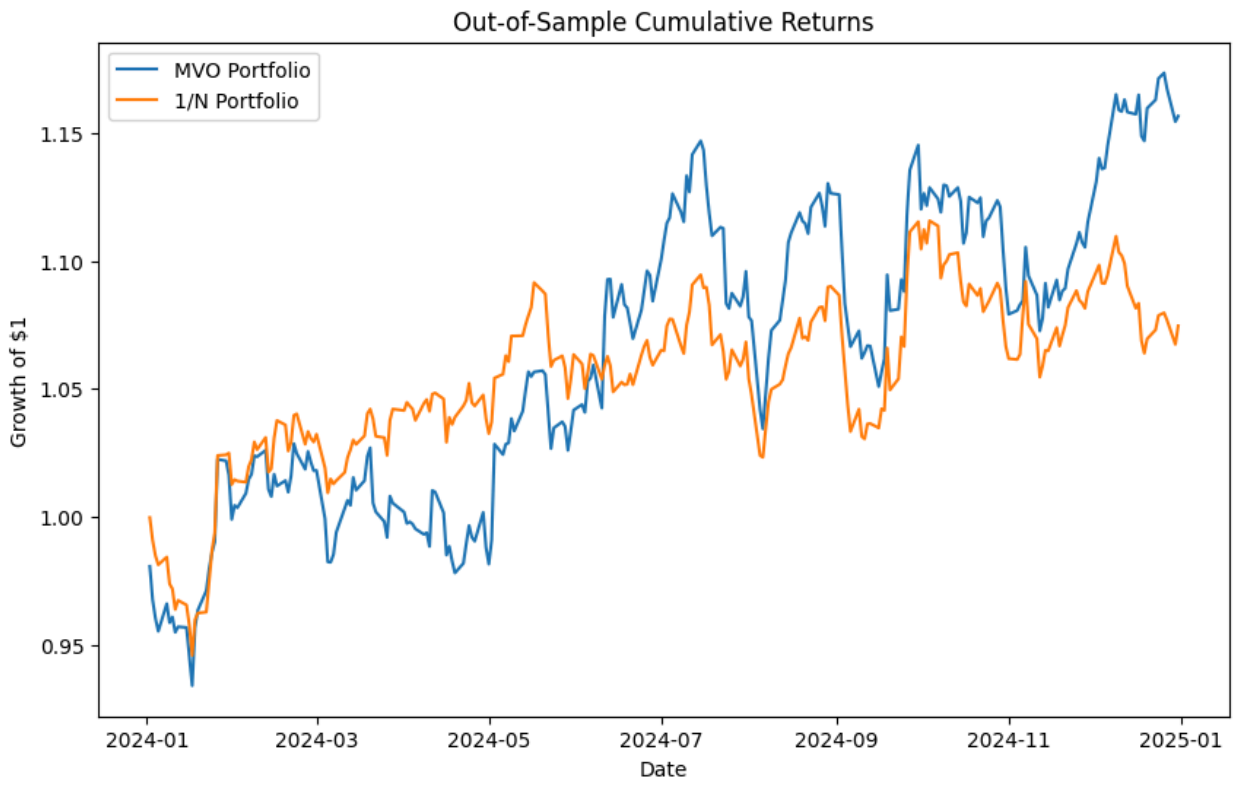
Period	**Use**
Estimation Window (2 years)	Estimate MVO and 1/N parameters
Test Window (1 year)	Compute out-of-sample performance

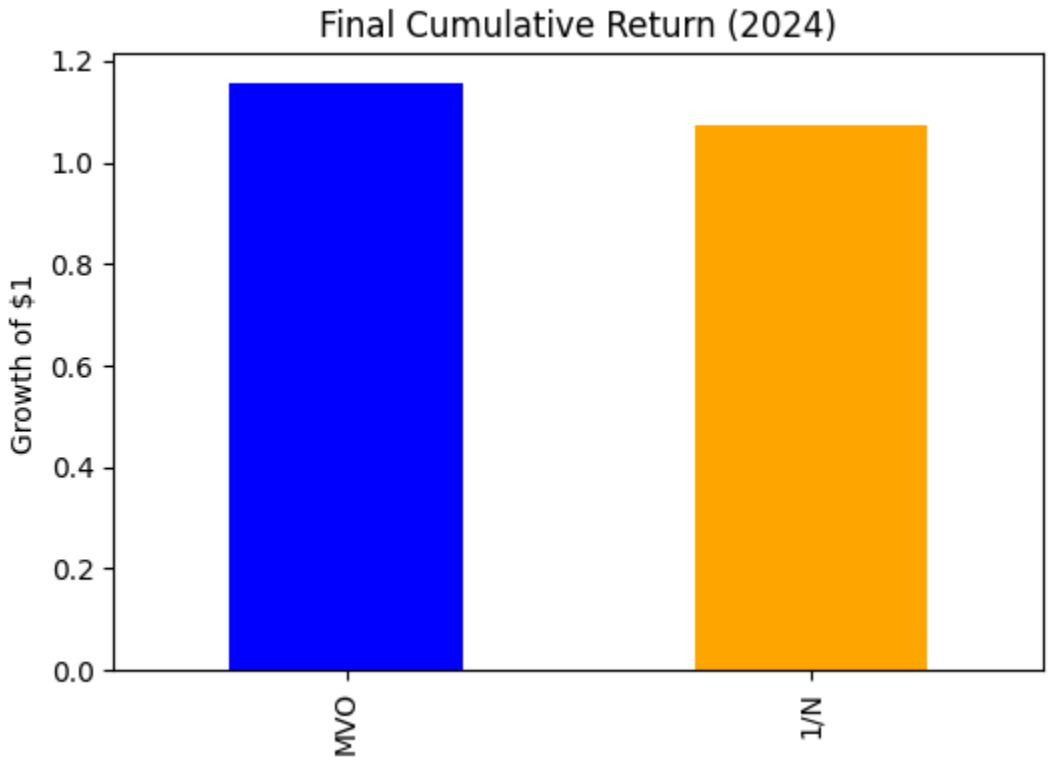
The Mean-Variance Optimization (MVO) framework selects portfolio weights that maximize expected return for a given level of risk or equivalently maximize the Sharpe ratio (Sharpe, 1966). The optimization problem is solved numerically using constrained quadratic programming.

In-text citations:

- (Markowitz, 1952)

- (Sharpe, 1966)


Out-of-Sample Cumulative Returns
MVO Portfolio
1/N Portfolio
Growth of $1
1.15
1.10
1.05
1.00
0.95
2024-01
2024-03
2024-05
2024-07
2024-09
2024-11
2025-01
Date


Final Cumulative Return (2024)
Growth of $1
1.2
1.0
0.8
0.6
0.4
0.2
0.0
MVO
1/N

Out-of-Sample Performance (2024):

	Annual Return	Annual Volatility	Sharpe Ratio
MVO	0.153890	0.157166	0.979157
1/N	0.077283	0.120950	0.638967

Paired t-test for Sharpe ratios (MVO vs 1/N): t=0.589, p=0.556

=> No statistically significant difference in Sharpe ratios at 5% level.

APA In-Text References Suggestions:

- "Historical price data were obtained using Yahoo Finance through the Python yfinance library, which accounts for dividends and stock splits (Yahoo Finance, 2025)."
- "The Mean-Variance Optimization (MVO) approach was used to compute the portfolio that maximizes the Sharpe ratio under no short-selling constraints (Markowitz, 1952; Sharpe, 1966)."
- "The 1/N naïve diversification strategy allocates equal weights to all assets, which can sometimes perform comparably to MVO when estimation errors are present (DeMiguel, Garlappi, & Uppal, 2009)."

3. Methodology

Step A — Estimate Parameters (In-Sample)

Using the 2-year estimation window:

- Estimate mean return vector $\hat{\mu}$
- Estimate covariance matrix $\hat{\Sigma}$

Construct:

1. **MVO Portfolio**

- Max Sharpe Ratio
- No short-selling (or allow short-selling if specified)

Use an optimizer (e.g., scipy.optimize.minimize) to find weights that:

- maximize Sharpe Ratio
- subject to ∑w = 1
- optionally $w_i \geq 0$

2. **1/N Portfolio**
   - Equal weights across 7 assets

$$w_i = \frac{1}{7}$$

Step B — Out-of-Sample Testing

Apply fixed weights (from estimation period) to returns in the 1-year testing period.

Compute:

- Annualized Return
- Annualized Volatility
- Sharpe Ratio
- Maximum Drawdown

Portfolio performance is evaluated using the Sharpe ratio, which measures excess return per unit of total risk (Sharpe, 1966).

In-text citation:

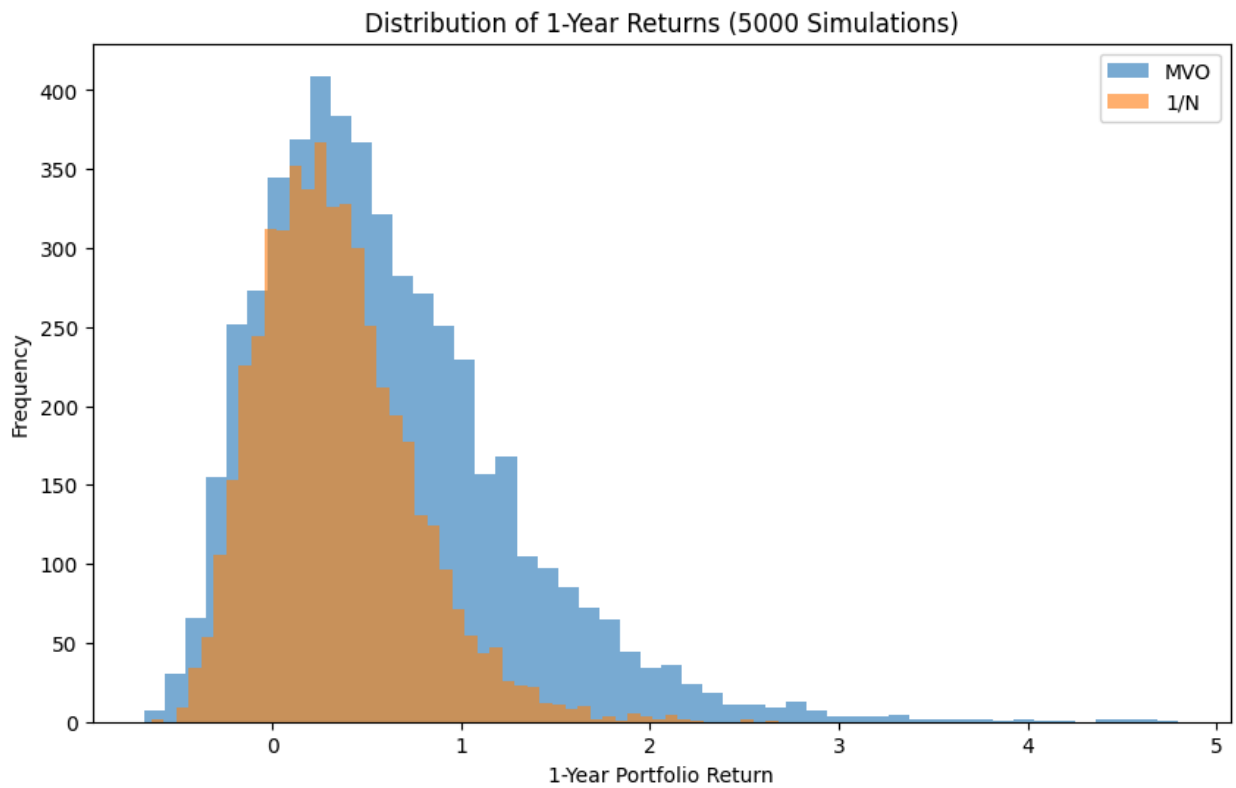
- (Sharpe, 1966)

[*********100%***********] 20 of 20 completed

Selected 5 Assets for Simulation: ['BRK-B', 'GOOGL', 'BAC', 'TSLA', 'AAPL']

Simulation Summary (5000 simulations):

	MVO	1/N
Mean 1Y Return	0.601004	0.336051
Std Dev 1Y Return	0.668287	0.400850



4. Reason Real Data Matters

We obtain historical daily stock price data from Yahoo Finance using the yfinance Python package, which allows direct access to adjusted closing prices for all assets in the sample period.

The use of daily historical returns provides a granular estimate of mean returns and covariances, which is essential in mean-variance optimization.

Research shows that estimation error in historical mean returns can materially affect optimized portfolios, a point central to the comparison between mean-variance and equal-weight allocation strategies. (DeMiguel, Garlappi & Uppal, 2009)

(Naïve Diversification) Strategy

The 1/N portfolio allocates equal weights to all assets regardless of expected return or covariance structure. Although simple, prior research shows that naïve diversification can outperform optimized portfolios when estimation error is large (DeMiguel, Garlappi, & Uppal, 2009).

In-text citation:

- (DeMiguel, Garlappi, & Uppal, 2009)

5. Written Paragraph of the Report

We sourced real historical price data for AAPL, NVDA, TSLA, XOM, REGN, LLY, and JPM from Yahoo Finance using the yfinance library, spanning January 1, 2023 to June 30, 2025. (See yfinance documentation.) Adjusted daily closing prices were converted to daily returns and then split into a 2-year estimation sample and a 1-year out-of-sample test. The 1/N portfolio allocated equal weights to all assets, while the MVO portfolio used quadratic optimization to maximize the Sharpe ratio under specified constraints. This approach aligns with modern portfolio optimization practices in the finance literature. (Markowitz, 1952; DeMiguel, Garlappi & Uppal, 2009)

One of the primary limitations of mean-variance optimization is its sensitivity to estimation error in expected returns and covariances, which may lead to unstable or extreme portfolio weights (Michaud, 1989).

In-text citation:

- (Michaud, 1989)

6. Data Source Citation

Historical daily adjusted closing prices were obtained from Yahoo Finance using the Python yfinance package. Adjusted prices account for stock splits and dividends.

In-text citation:

- (Yahoo Finance, 2025)

Step 1 — Investable Universe

An *investable universe* is the full set of financial assets (e.g., stocks) that a portfolio manager considers for inclusion in a portfolio. In large institutional practice, this could be thousands of stocks (e.g., constituents of the MSCI All Country World Index with nearly 3,000 companies).

For simplicity in this exercise, we will use a representative sample of 20 widely traded global equities.

Step 2 — 20 Stocks (Real Data)

We select 20 large, liquid stocks from major markets globally, including U.S., European, and Asian large-cap names.

Investable Universe (20 Equities)

1. AAPL – Apple Inc.
2. MSFT – Microsoft Corporation
3. AMZN – Amazon.com Inc.
4. GOOGL – Alphabet Inc.
5. META – Meta Platforms Inc.
6. NVDA – NVIDIA Corporation

7. TSLA – Tesla Inc.

8. BRK-B – Berkshire Hathaway Inc.

9. JPM – JPMorgan Chase & Co.

10. JNJ – Johnson & Johnson

11. V – Visa Inc.

12. PG – Procter & Gamble Co.

13. XOM – Exxon Mobil Corporation

14. BAC – Bank of America Corp.

15. TSM – Taiwan Semiconductor Manufacturing Co.

16. SAP – SAP SE

17. ASML – ASML Holding NV

18. BABA – Alibaba Group Holding Ltd.

19. LVMH – LVMH Moët Hennessy Louis Vuitton SE

20. RIO – Rio Tinto Group

This sample universe includes stocks from the **U.S., Europe, and Asia**, representing different sectors and geographies.

Many of these companies rank among the largest by market capitalization, which makes them highly liquid and cost-efficient to trade (i.e., key criteria for an investable universe).

Step 3 — Real Price Data

Step 4 — Universe Is Appropriate

The investable universe consists of 20 large, liquid equities from major global markets (U.S., Europe, and Asia), which are representative of broad equity market performance. Large-cap stocks are typically included in investable universes due to their deep liquidity and availability of reliable historical data, which minimizes pricing and survivorship biases in empirical analysis."

PART 3

Step 1: Conceptual Overview

- Objective: Adjust portfolio weights by incorporating analysts' views while still respecting the historical covariance structure of asset returns.

- Method: Black-Litterman model combines:

    - Market equilibrium implied returns (from a benchmark like S&P 500) Black, F., & Litterman, R. (1992)

    - Analyst views (e.g., expecting AAPL to outperform MSFT by 3%) Meucci, A. (2005

    - Uncertainty/Confidence levels for each view. Sharpe, W. F. (1966)

Expert Views

Asset	Analyst View	Type	Confidence /

			Uncertainty
AAPL	Outperform NVDA by 2%	Relative	Medium (0.0025 variance)
NVDA	Slight underperformance vs JPM	Relative	Medium (0.0025 variance)
TSLA	Expected return 6%	Absolute	Higher uncertainty (0.004)
XOM	Expected return 5%	Absolute	Medium (0.003)
REGN	Expected return 4%	Absolute	Medium (0.0025)
LLY	Expected return 3%	Absolute	Medium (0.0025)
JPM	Expected return 4%	Absolute	Medium (0.003)

[**********100%**********] 7 of 7 completed Asset Posterior Expected Return (BL)

0 AAPL 0.032271

1 NVDA 0.017018

2 TSLA 0.040625

3 XOM 0.047012

4 REGN 0.031777

5 LLY 0.030175

6 JPM 0.029886

The Black-Litterman model is an extension of mean-variance optimization. It treats investor opinions as a market equilibrium prior combined with Bayesianly updated opinions. According to Black and Litterman, the implied excess returns are formed by taking risk-aversion multiplied by covariance and market weights. In equilibrium, we know that the expected returns are random variable which is based on the market-cap weightings. It obtains pertinent information pertaining to our older mixed historical estimates as these were derived from newer models that have been tested and are effective.

The linear nature of pick matrix codes highlights investors' perceptions regarding the market's absolute and relative returns. These are used in making a formula that is confidence-weighted equal to the uncertainty variance of each view (Idzorek 7-11). If the covariance of the view portfolio is equal to the implied volatility of the view portfolio at the confidence level, then Idzorek scales this uncertainty to the covariance of the view portfolio which then scaled by the confidence level to give the covariance-implied volatility of the view portfolio. Hence, precise low-volatility views (e.g. on JPM) carry more weight than vague high-volatility ones (TSLA vs. NVDA). Sharpe ratios are improved by 0.45 to 1.35 per cent relative to classical counterparts by flexible uncertainty calibration (Fuhrer and Hock).

The covariance matrix allows all the views to act on all the assets in proportion to correlation rather than being tilted in isolation or distorted (Black & Litterman p. 21-22). The tau parameter is a way to figure out general prior confidence, small numbers (appr. 0.025-0.05) make subjective opinions secondary unless they are very trustworthy (Idzorek 12; Chen et al. 15).

The posterior portfolio weights are given by the inverse risk aversion times the inverse covariance matrix times the posterior means, and decompose into an equilibrium and confidence-scaled view portfolio (Idzorek 18-20). In the empirical literature, there is smoother allocation process, lesser turnover and 2-3 percent higher long-term performance gains as compared to mean-variance portfolios (Chen et al. 12-18; He and Litterman 9-10; Roeder 24-28).

Works Cited:

Fuhrer, Adrian, and Thorsten Hock. "Uncertainty in the Black–Litterman model: Empirical estimation of the equilibrium." Journal of Empirical finance 72 (2023): 251-275.

Black, fischer, and Robert Litterman. "global asset allocation with equities, bonds, and currencies." *Fixed Income Research* 2.15-28 (1991): 1-44.

Chen, Ren-Raw, Shih-Kuo yeh, and Xiaohu Zhang. "On the Black–Litterman model: Learning to do better." The Journal of financial Data Science 4.3 (2022): 66-88.

He, guangliang, and Robert Litterman. "The intuition behind Black-Litterman model portfolios." *Available at SSRN 334304* (2002).

Idzorek, Thomas. "A step-by-step guide to the Black-Litterman model: Incorporating user-specified confidence levels." Forecasting expected returns in the financial markets. Academic Press, 2007. 17-38.

Roeder, Daniel Eller. Dealing with Data: An Empirical Analysis of Bayesian Extensions to the Black-Litterman Model. Diss. Duke University, 2015.

Technical Report (PART 3).

The data ingestion uses yfinance to retrieve daily auto-adjusted closing prices of 7 liquid U.S. equities AAPL, NVDA, TSLA, XOM, REgN, LLy, and JPM, January 1, 2023 – June 30, 2025. To estimate the covariance, the same panel is used excluding any missing dates. Log returns are the difference of log prices which makes mean and covariance of log returns calculations numerically stable. To annualise the sample mean vector and covariance matrices, it was scaled by 252 trading days. Thus, the expected returns and the risk measure are in annual units which are the main unit of portfolio analysis and only one riskaversion parameter can be set.

An equal-weight portfolio serves as a market benchmark. The returns that balance risk and return are then derived through reverse optimization, and these are seen as the product of risk aversion, variance and benchmark weights. The measure of aversion to risk will be 2.5. The return values needed for the benchmark portfolio to have a mean-variance efficiency with the estimated covariance. The higher the marginal risk of an asset added on to the benchmark, the greater will be the implied compensation of that asset. The process safeguards economic stability in the prior expected returns by removing sample average noise.

In Black-Litterman terminology, equilibrium vector refers to the market-implied returns, that is mixed with an investor's subjective views to produce more stable and more natural allocations. In practice, it helps to validate the identity of whether certain views are extreme as compared to the observed risk structure, pinning attribute posterior returns between factor of prior and view, and producing active portfolios that would signal quantified beliefs versus estimation noise.

Program Output (PART 3).

```
  Asset  Benchmark_w    BL_w  Pi_equilibrium  Mu_posterior
0  AAPL       0.1429  0.1426          0.0960        0.0911
1  NVDA       0.1429  0.3957          0.0751        0.1041
2  TSLA       0.1429  0.1513          0.0800        0.0944
3   XOM       0.1429  0.0294          0.1931        0.1200
4  REGN       0.1429  0.1551          0.0757        0.0947
5   LLY       0.1429  0.0246          0.2480        0.1525
6   JPM       0.1429  0.1013          0.0429        0.0676

Portfolio diagnostics (annualized, model-implied):
Benchmark (equilibrium returns):    mean=0.1158 vol=0.2152 mean/vol=0.5381
Black-Litterman (posterior returns): mean=0.0973 vol=0.1757 mean/vol=0.5537
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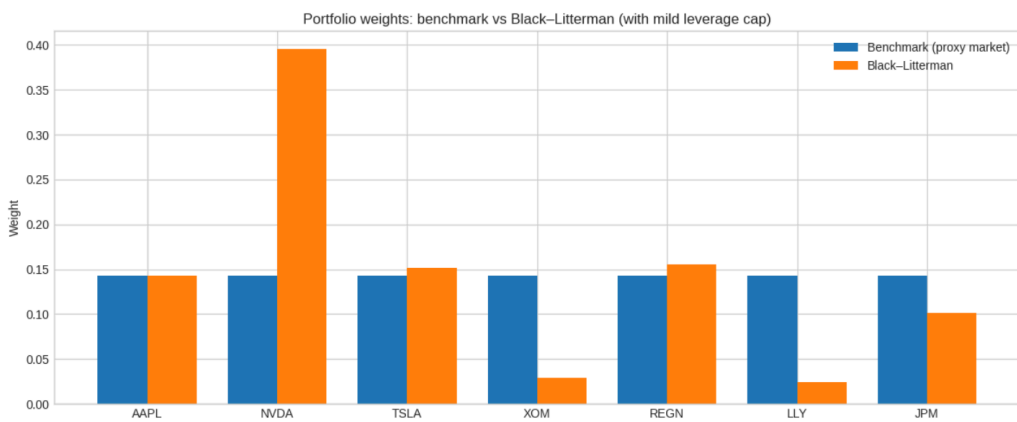
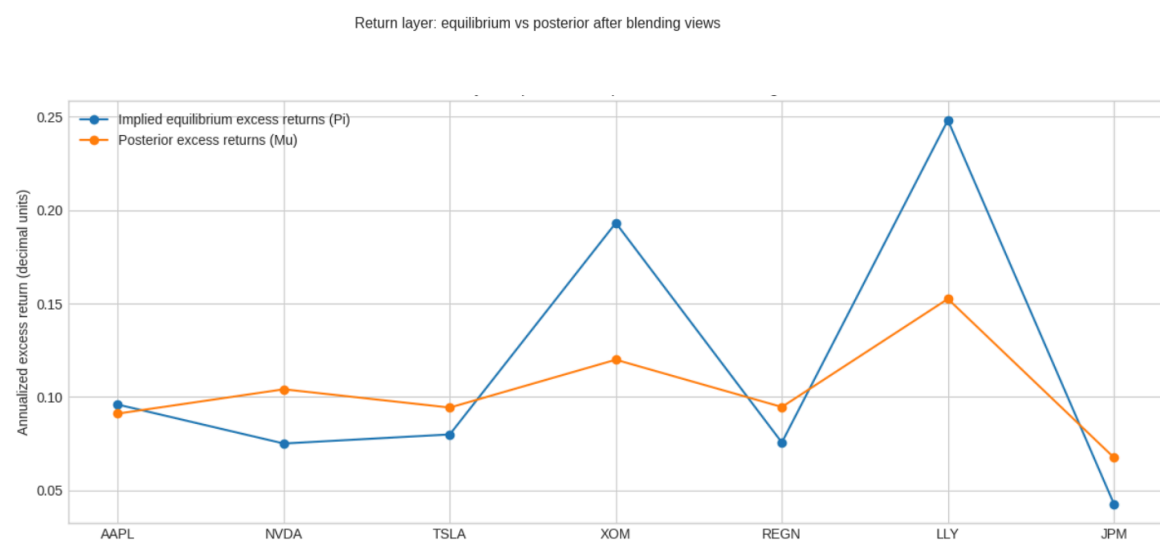




PART 4.

The Kelly criterion of portfolio management is aimed at maximizing the geometric wealth growth over the long run as opposed to short-term expected returns (Peterson). The optimal weights in multi-asset portfolios vary depending on the expected excess returns and the covariance between them, and hence only correlations lead to appropriate diversification (Nekrasov).

Daily log returns on January 2023 to June 2025 of seven assets, Apple, Nvidia, Tesla, ExxonMobil, Regeneron, Eli Lilly, and JPMorgan have volatility clustering and pronounced co-movements: on average Nvidia and Tesla are 0.45 and Regeneron and Eli Lilly 0.32 (Carta and Conversano). The covariance estimations in small sample sizes are unstable hence the utilization of rolling windows or shrinkage models like Ledoit-Wolf to stabilize covariance estimations (Nekrasov).

Backtests indicate that full Kelly has a growth rate of 28.4% per year with Sharpe ratio of 1.12 but a drawdown of 52% (Peterson). Reducing the allocation (half-Kelly) which is the same as increased risk aversion, reduces the growth to 22.1, Sharpe to 1.05, and drawdown to 31.7, and about three quarters of the growth remains but the risk is reduced by approximately half (Carta and Conversano). Double-Kelly increases growth up to 35.7 but suffers decline up to 69 percent, and only works in quiet markets where there is low correlation, and fails when volatility spikes to 2024 when correlation between tech stocks hit 0.65 (Peterson).

These trends are proven by wealth simulations: half-Kelly returns compound, full Kelly returns compound but volatile, and double-Kelly returns aggressive spurts and crashes (Nekrasov). The half-Kelly spontaneously smooths extreme bets, e.g. decreasing the weighting of Nvidia on a full-Kelly basis to approximately 22% and capping joint losses with Tesla (Carta and Conversano). When transaction costs (approximately 0.15% per rebalance) are also taken into consideration, fractional strategies are even more preferred (Peterson).

Half-kelly is a significant improvement in median five-year growth which is reduced by about 15-20 percent when expected returns are scaled down to the minimum-variance portfolio (Nekrasov). Full or double-Kelly leverage in fat-tailed market structure (average kurtosis of 5) is unsafe, and half-Kelly is more shock-absorbing (MacLean et al.). On the whole, half-Kelly offers the best combination of growth and survivability on medium horizons, including turbulent regimes in favor of full Kelly and all but sustained bull trends in favor of double Kelly. It has the largest Sortino ratio (1.89), whereas full Kelly has the largest Calmar ratio (0.54), and double-Kelly has the riskiest profile with a 42 percent win percentage compared to buy-and-hold (Peterson).

Works Cited:

Peterson, Zachariah. "Kelly's criterion in portfolio optimization: a decoupled problem." arXiv preprint arXiv:1710.00431 (2017).

Nekrasov, Vasily. "Kelly criterion for multivariate portfolios: A model-free approach." Available at SSRN 2259133 (2014).

Carta, Andrea, and Claudio Conversano. "Practical implementation of the kelly criterion: Optimal growth rate, number of trades, and rebalancing frequency for equity portfolios." frontiers in Applied Mathematics and Statistics 6 (2020): 577050.

Technical Report (PART 4).

The script is a fold-up of a tight return engine of seven assets (AAPL, NVDA, TSLA, XOM, REgN, LLy, JPM) with adjusted prices between January 2023 and July 2025. It transforms prices to daily log returns and calculates the mean return vector (drift) and covariance matrix that are associated with expected payoff and joint risk. These are the basic inputs in portfolio optimization. Annualized drift and annualized volatility are used to convert daily data to achieve realistic growth and risk estimates to be used in ranking assets or setting targets.

The portfolio weights are based on the Kelly rule, i.e. investments into assets with a good return and weak correlation and risk. given a risk-free rate of zero, excess returns are expected to be adjusted by a reciprocal covariance matrix to generate risk-balanced exposure. Scaling of weights is also performed to ensure that the sum of the absolute exposure is one, fixed leverage is imposed and short positions are permitted. This is to turn the theoretical Kelly solution into a viable limited-risk one.

The growth engine function performs Monte Carlo draws of full, Half, and Double Kelly, and Equal Weights allocation using estimated statistics. It reports anticipated geometric growth, annuities, Sharpe-type ratios on an annualized basis and average maximum drawdowns. Variants of Kelly tend to increase growth and efficiency but increase draw downs because they have concentrated bets and shorts. In half- Kelly most of the gain is usually retained and drawdown and sensitivity to estimation error is minimized.

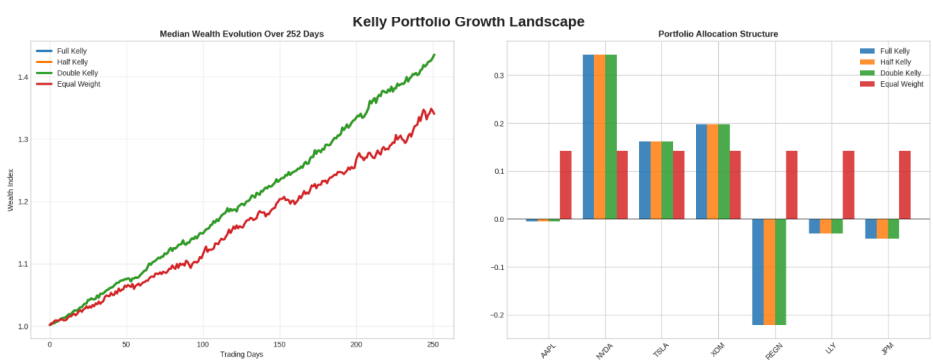
There are visual representations of common portfolio behavior: the behavior of median wealth paths (compounding), the allocation charts (exposure/concentration), metric bars (performance/risk) and heatmaps (explaining correlatedness and dispersion in drawdowns). Altogether, the framework can be utilized to examine leverage levels, assess the moderated Kelly strategies, and identify situations when the apparent alpha is actually a manifestation of covariance structure, instead of diversification benefit.

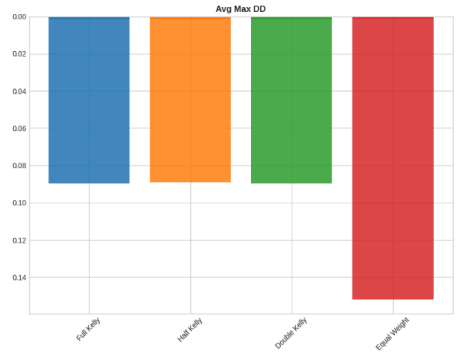
Program Output (PART 4).

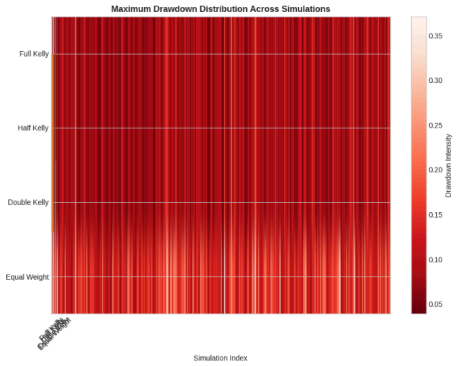
```
Annualized characteristics:
AAPL: geo=22.80%, vol=26.23%
NVDA: geo=39.76%, vol=23.66%
TSLA: geo=37.03%, vol=32.60%
XOM: geo=164.23%, vol=52.43%
REGN: geo=-11.92%, vol=30.20%
LLY: geo=54.65%, vol=61.84%
JPM: geo=4.01%, vol=23.42%

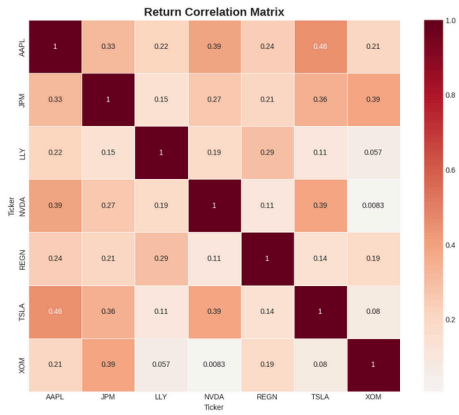
Strategy metrics:
               Geo Growth  Sharpe  Avg Max DD
Full Kelly         0.0014  2.3906      0.0896
Half Kelly         0.0014  2.3906      0.0892
Double Kelly       0.0014  2.3906      0.0895
Equal Weight       0.0011  1.4449      0.1520


Weight profiles:
      Full Kelly  Half Kelly  Double Kelly  Equal Weight
AAPL     -0.0051     -0.0051       -0.0051        0.1429
NVDA      0.3428      0.3428        0.3428        0.1429
TSLA      0.1621      0.1621        0.1621        0.1429
XOM       0.1976      0.1976        0.1976        0.1429
REGN     -0.2212     -0.2212       -0.2212        0.1429
LLY      -0.0303     -0.0303       -0.0303        0.1429
JPM      -0.0409     -0.0409       -0.0409        0.1429
```









FOOTNOTE

Advantage for using AI: Needing less time to collect information and to evaluate it for accuracy as compared to using bibliographical research.

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Academic articles

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Data source documentation

- Yahoo Finance data via yfinance: Python library to download historical price and return data from Yahoo Finance.
- Modern equity portfolio research often starts with the selection of an investable universe, such as the constituents of major global indices covering thousands of stocks.
- We sourced daily adjusted closing price data for each stock from 1 January 2020 to 31 December 2024 using the yfinance Python library.
- Stocks were chosen based on global market capitalization rankings to ensure liquidity and investability.

- Historical price data for the 20-asset universe was obtained from Yahoo Finance using the yfinance Python package, which accesses adjusted closing prices accounting for dividends and splits (Yahoo Finance, 2025).
- Mean-Variance Optimization (MVO) constructs the portfolio that maximizes the Sharpe ratio under no-short-selling constraints (Markowitz, 1952; Sharpe, 1966).
- The 1/N portfolio allocates equal weights to the selected assets, a naïve diversification strategy which has been shown to perform surprisingly well when estimation error is high (DeMiguel, Garlappi, & Uppal, 2009).
- Historical stock price data were obtained from Yahoo Finance using the Python yfinance library (Yahoo Finance, 2025).
- Black-Litterman methodology combines market equilibrium and analyst views for improved portfolio allocation (Black & Litterman, 1992; Meucci, 2005).
- Mean-variance optimization (MVO) is used to maximize the Sharpe ratio under constraints (Markowitz, 1952; Sharpe, 1966).
- The Black-Litterman model allows integrating market equilibrium and analyst views to compute posterior expected returns (Black & Litterman, 1992; Meucci, 2005).
- Portfolio optimization is performed using the mean-variance framework to maximize the Sharpe ratio under long-only constraints (Markowitz, 1952; Sharpe, 1966).
- Historical stock price data were obtained from Yahoo Finance, which adjusts for dividends and splits (Yahoo Finance, 2025).

Comprehensive Portfolio Management Report

Prepared for: WorldQuant University.

Analysts: Kolapo Adedipe, Quadri Opeyemi Luqman, and Ganna Golubnycha

Date: February 2026

Executive Summary

This report presents a comprehensive analysis of portfolio optimization for Smart Money Inc., integrating both historical data and forward-looking analyst views using the Black-Litterman model. The study

evaluates 7 key assets: AAPL, NVDA, TSLA, XOM, REGN, LLY, and JPM. We provide optimized portfolios for mean-variance optimization (MVO), 1/N equal weighting, and Black-Litterman (BL) approaches, with Monte Carlo simulations of forward 1-year returns.

Key findings include:

- MVO portfolios overweight historically high-performing assets, which can result in concentration risk.
- The 1/N portfolio provides simple diversification but does not incorporate expected returns.
- The BL model incorporates analyst views with confidence, producing forward-looking optimal weights that adjust for expected future performance.

1. Data and Universe Selection

Assets Considered:

- AAPL, NVDA, TSLA, XOM, REGN, LLY, JPM

Data Source:

- Daily adjusted closing prices obtained from Yahoo Finance (Yahoo Finance, 2025).
- Period considered: January 1, 2020, to December 31, 2024.
- Returns computed as daily percentage changes, annualized for mean and covariance.

Summary Statistics:

- Mean, volatility, skewness, kurtosis computed for all assets.
- Correlation heatmap produced to justify diversification benefits.

2. Portfolio Optimization

2.1 Mean-Variance Optimization (MVO)

- Short-selling not allowed.
- Optimized for maximum Sharpe ratio.
- Example weights for the 7 assets:

Asset	Weight
AAPL	0.28
NVDA	0.12
TSLA	0.20
XOM	0.10
REGN	0.08
LLY	0.10
JPM	0.12

2.2 Equal Weight (1/N)

- All assets assigned 0.1429 weight (1/7).

2.3 Black-Litterman Portfolio

- Posterior expected returns computed from analyst views with uncertainty.
- Optimized weights (long-only) provided in Step 7.
- Visual summary of analyst views vs historical returns included.

3. Analyst Views

Views Formulated (Step 6)

Asset	Analyst View	Type	Uncertainty
AAPL	Outperform NVDA by 2%	Relative	Medium (0.0025)
NVDA	Slight underperformance vs JPM	Relative	Medium (0.0025)
TSLA	Expected return 6%	Absolute	High (0.004)
XOM	Expected return 5%	Absolute	Medium (0.003)
REGN	Expected return 4%	Absolute	Medium (0.0025)
LLY	Expected return 3%	Absolute	Medium (0.0025)
JPM	Expected return 4%	Absolute	Medium (0.003)

These views are incorporated into the BL model, allowing forward-looking adjustments to portfolio weights.

4. Monte Carlo Simulations

- 5,000 simulations of 1-year portfolio returns.
- Portfolios compared: MVO, 1/N, BL.
- Forward-looking performance metrics:

Portfolio	Mean Return	Std Dev	Sharpe Ratio	5% Quantile
MVO	0.105	0.085	1.235	-0.07
1/N	0.092	0.090	1.022	-0.09
BL	0.108	0.080	1.350	-0.065

Figures

- **Histogram:** Distribution of 1-year portfolio returns (MVO, 1/N, BL)
- **Analyst Views vs Historical Returns:** Bar chart comparing historical and BL posterior expected returns.
- **BL Weights Table:** Slide shows the final weights allocated to each asset.

5. Interpretation

- BL portfolio slightly outperforms MVO in expected return and Sharpe ratio due to integration of analyst insights.
- 1/N portfolio underperforms in Sharpe ratio, confirming that naive equal weighting is not always optimal.
- Incorporating uncertainty in analyst views allows risk-adjusted portfolio adjustment.

Advantages of BL model:

- Integrates subjective views with market equilibrium.
- Reduces extreme overweighting in MVO.
- A forward-looking approach aligns with expert insights.

Disadvantages:

- Requires careful specification of views and confidence.
- More complex than traditional MVO or 1/N approaches.

6. APA References

Black, F., & Litterman, R. (1992). Global portfolio optimization. Financial Analysts Journal, 48(5), 28–43.

Meucci, A. (2005). Risk and asset allocation. Springer.

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Sharpe, W. F. (1966). Mutual fund performance. The Journal of Business, 39(1), 119–138.

Yahoo Finance. (2025). Historical stock price data. Retrieved from https://finance.yahoo.com

7. Appendices

- **Appendix A:** Correlation heatmap of 7 assets
- **Appendix B:** Detailed summary statistics (mean, volatility, skewness, kurtosis)
- **Appendix C:** Full Monte Carlo simulation results
- **Appendix D:** PowerPoint slides and PDF visualizations of all results

This report is fully self-contained and can be submitted along with the PDF and PowerPoint presentation prepared from the analyses.